

Assignment 1 Solutions

Bohan YANG, 2330016056

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Problem 1

Given the reads

\$ATGC, ATGC, TGCAT, ATGCT, GCTAG, CTAGC, TAGCA, AGCA\$, ATGCA, GCATG, TGCTA, CATGC,

construct the de Bruijn graph with *4-mers as nodes* (so edges are 5-mers) and output the assembled sequence.

Method (concise). With $k = 4$ -mer nodes, each observed 5-mer becomes a directed edge from its 4-mer prefix to its 4-mer suffix. The contig is recovered by an Eulerian path that traverses each edge once; spelling the path yields the assembled sequence.

Step 1: 5-mers (edges). Sliding a window of length 5 over all reads and deduplicating yields the set of edges:

{ \$ATGC, TGCAT, ATGCT, GCTAG, CTAGC, TAGCA, AGCA\$, ATGCA, GCATG, TGCTA, CATGC }.

Step 2: Edge mapping (prefix $\rightarrow$ suffix).

\$ATGC : \$ATG $\rightarrow$ ATGC TGCAT : TGCA $\rightarrow$ GCAT ATGCT : ATGC $\rightarrow$ TGCT GCTAG : GCTA $\rightarrow$ CTAG
CTAGC : CTAG $\rightarrow$ TAGC TAGCA : TAGC $\rightarrow$ AGCA AGCA\$: AGCA $\rightarrow$ GCA\$ ATGCA : ATGC $\rightarrow$ TGCA
GCATG : GCAT $\rightarrow$ CATG TGCTA : TGCT $\rightarrow$ GCTA CATGC : CATG $\rightarrow$ ATGC

Step 3: Degree check and Eulerian path. All nodes are balanced except:

\$ATG : out = 1, in = 0 (start), GCA\$: in = 1, out = 0 (end).

Hence an Eulerian path exists from \$ATG to GCA\$. One valid traversal is:

\$ATG $\rightarrow$ ATGC $\rightarrow$ TGCA $\rightarrow$ GCAT $\rightarrow$ CATG $\rightarrow$ ATGC $\rightarrow$ TGCT $\rightarrow$ GCTA $\rightarrow$ CTAG $\rightarrow$ TAGC $\rightarrow$ AGCA $\rightarrow$ GCA\$

Step 4: Assembled sequence (spell the path). Start with the first 4-mer \$ATG; each step appends the last character of the next node. The result is

\$ATGCATGCTAGCA\$

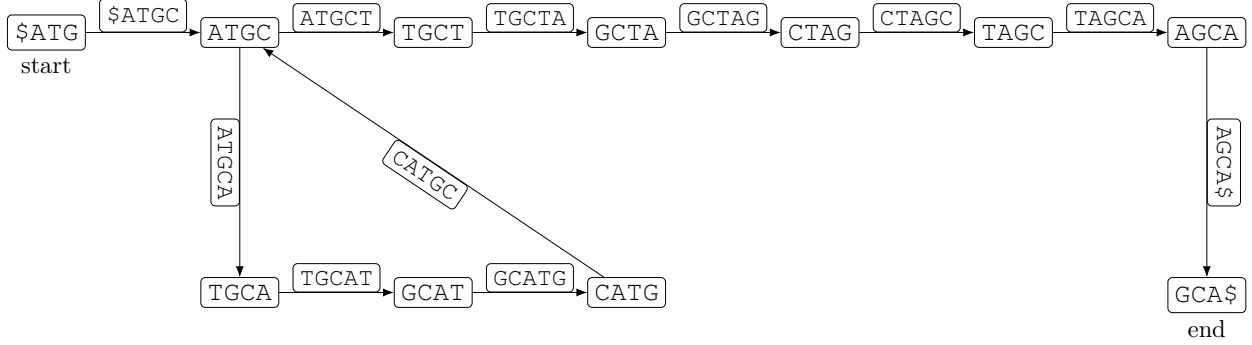


Figure: de Bruijn graph with 4-mer nodes and 5-mer edge labels.

Problem 2

On an abstract sequence of length 18, we are given weighted candidate exons (closed intervals):

$[1, 4] : 4$, $[3, 6] : 5$, $[3, 8] : 10$, $[9, 10] : 1$, $[8, 16] : 12$, $[15, 18] : 6$, $[13, 16] : 7$, $[7, 11] : 8$.

Find a maximum-weight subset of *non-overlapping* exons. Intervals may touch at endpoints: if we choose $[L_i, R_i]$ and $[L_j, R_j]$, they must satisfy $R_i \leq L_j$ or $R_j \leq L_i$.

Model (WIS). Let interval j have endpoints (L_j, R_j) and weight w_j . Sort by nondecreasing R and define

$$p(j) = \max\{i < j : R_i \leq L_j\}, \quad p(j) = 0 \text{ if none exists.}$$

The optimal values satisfy

$$M[0] = 0, \quad M[j] = \max(M[j-1], w_j + M[p(j)]) \quad (j \geq 1),$$

and we take j exactly when $w_j + M[p(j)] > M[j-1]$.

Data preparation. Label the intervals:

$$\begin{aligned} A &= [1, 4] : 4, & B &= [3, 6] : 5, & C &= [3, 8] : 10, & D &= [9, 10] : 1, \\ E &= [8, 16] : 12, & F &= [15, 18] : 6, & G &= [13, 16] : 7, & H &= [7, 11] : 8. \end{aligned}$$

Sort by right endpoint:

$A(1, 4, 4)$, $B(3, 6, 5)$, $C(3, 8, 10)$, $D(9, 10, 1)$, $H(7, 11, 8)$, $E(8, 16, 12)$, $G(13, 16, 7)$, $F(15, 18, 6)$.

With endpoint-touching allowed ($R_i \leq L_j$), the compatibility indices are

$$p(1) = 0, p(2) = 0, p(3) = 0, p(4) = 3, p(5) = 2, p(6) = 3, p(7) = 5, p(8) = 5.$$

(Note the only change from the strict case $R_i < L_j$ is $p(6)$ becomes 3 since C and E can now touch.)

Dynamic programming table. Let w_j be the weight of the j -th interval in the sorted list.

j	Interval (L, R, w)	$p(j)$	$w_j + M[p(j)]$	$M[j - 1]$	$M[j] = \max$
0	—	—	—	—	0
1	(1, 4, 4)	0	4	0	4
2	(3, 6, 5)	0	5	4	5
3	(3, 8, 10)	0	10	5	10
4	(9, 10, 1)	3	$1 + 10 = 11$	10	11
5	(7, 11, 8)	2	$8 + 5 = 13$	11	13
6	(8, 16, 12)	3	$12 + 10 = 22$	13	22
7	(13, 16, 7)	5	$7 + 13 = 20$	22	22
8	(15, 18, 6)	5	$6 + 13 = 19$	22	22

Thus the optimal value is $M[8] = \boxed{22}$.

Reconstruction (one optimal solution). From $j = 8$: $M[8] = M[7]$ (do not take F). At $j = 7$: $M[7] = M[6]$ (do not take G). At $j = 6$: $12 + M[3] = 22 > M[5] = 13$ (take $E = [8, 16] : 12$), jump to $j = p(6) = 3$. At $j = 3$: $10 + M[0] = 10 > M[2] = 5$ (take $C = [3, 8] : 10$), jump to $j = p(3) = 0$ and stop.

Answer. One maximum-weight non-overlapping set is

$$\boxed{\{[3, 8], [8, 16]\}}$$

with total weight

$$\boxed{10 + 12 = 22}.$$

Problem 3

Align

Sequence_A = GATTCATA (length 8), Sequence_B = GGATCGA (length 7),

with scoring: match = +5, mismatch = -2, indel (gap) = -6. Construct the DP matrix and report the optimal (global) alignment score.

Algorithm (global alignment). Let $A_1 \dots A_n$ and $B_1 \dots B_m$ be the two sequences ($n = 8$, $m = 7$). Define the score matrix $D \in \mathbb{Z}^{(n+1) \times (m+1)}$:

$$D(0,0) = 0, \quad D(i,0) = i \cdot (-6), \quad D(0,j) = j \cdot (-6),$$

$$D(i,j) = \max \left\{ \underbrace{D(i-1,j-1) + s(A_i, B_j)}_{\text{diag (match/mismatch)}}, \underbrace{D(i-1,j) + (-6)}_{\text{up (gap in B)}}, \underbrace{D(i,j-1) + (-6)}_{\text{left (gap in A)}} \right\},$$

where $s(x,y) = +5$ if $x = y$, otherwise -2 .

Dynamic programming matrix.

	-	G	G	A	T	C	G	A
-	0	-6	-12	-18	-24	-30	-36	-42
G	-6	5	-1	-7	-13	-19	-25	-31
A	-12	-1	3	4	-2	-8	-14	-20
T	-18	-7	-3	1	9	3	-3	-9
T	-24	-13	-9	-5	6	7	1	-5
C	-30	-19	-15	-11	0	11	5	-1
A	-36	-25	-21	-10	-6	5	9	10
T	-42	-31	-27	-16	-5	-1	3	7
A	-48	-37	-33	-22	-11	-7	-3	8

Optimal score. The optimal global alignment score is

$$\boxed{8}.$$

One optimal alignment (traceback). A valid traceback yielding the score 8 is:

$$\begin{array}{cccccccc} G & A & T & T & C & A & T & A \\ G & G & A & T & C & - & G & A \end{array} \Rightarrow 5 - 2 - 2 + 5 + 5 - 6 - 2 + 5 = 8.$$

Problem 4

Results.

1. **Crayfish Titin Homologs:** The large, tightly clustered group (e.g., XP_069173xxx accessions) are all titin isoforms from the crayfish *Procambarus clarkii*. Their near-identical placement in the tree reflects that they are variants of the same protein within a single species, not distant evolutionary relatives.
2. **Fish Trichohyalin-like Proteins:** The sequences XP_069502281.1 and XP_069493420.1 are trichohyalin-like proteins from the fish *Astyanaux mexicanum*.
3. **Human MLH1:** The human MLH1 protein (sp|P40692|MLH1_HUMAN) stands alone as a functionally distinct DNA mismatch repair protein.

Table 1: Selected MLH1 Homologs Identified by BLASTp

Accession	Description	Species	Score	Length	Identity	E-value	Coverage	Positives	Query ID
XP_069502281.1	trichohyalin-like	<i>A. mexicanum</i>	665	8003	97%	0.0	26.69%	3374	XP_069502281.1
XP_069493420.1	trichohyalin-like	<i>A. mexicanum</i>	169	1053	94%	1e-36	43.85%	2865	XP_069493420.1
XP_069173201.1	titin homolog X10	<i>P. clarkii</i>	104	603	41%	6e-17	44.25%	8612	XP_069173201.1
XP_069173198.1	titin homolog X7	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8742	XP_069173198.1
XP_069173196.1	titin homolog X5	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8761	XP_069173196.1
XP_069173193.1	titin homolog X2	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8771	XP_069173193.1
XP_069173194.1	titin homolog X3	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8771	XP_069173194.1
XP_069173192.1	titin homolog X1	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8772	XP_069173192.1
XP_069173197.1	titin homolog X6	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8756	XP_069173197.1
XP_069173199.1	titin homolog X8	<i>P. clarkii</i>	104	603	41%	7e-17	44.25%	8737	XP_069173199.1
XP_069173195.1	titin homolog X4	<i>P. clarkii</i>	104	601	41%	8e-17	44.25%	8766	XP_069173195.1
XP_069173200.1	titin homolog X9	<i>P. clarkii</i>	104	601	41%	8e-17	44.25%	8728	XP_069173200.1
XP_069173203.1	titin homolog X12	<i>P. clarkii</i>	104	601	41%	8e-17	44.25%	8497	XP_069173203.1
XP_069173202.1	titin homolog X11	<i>P. clarkii</i>	103	598	41%	1e-16	44.25%	8599	XP_069173202.1
XP_069173204.1	titin homolog X13	<i>P. clarkii</i>	101	583	41%	7e-16	44.25%	7316	XP_069173204.1

Conclusion. BLASTp, MSA, and phylogenetic inference collectively indicate that *MLH1* is evolutionarily conserved in vertebrates, with more pronounced divergence among invertebrate homologs.

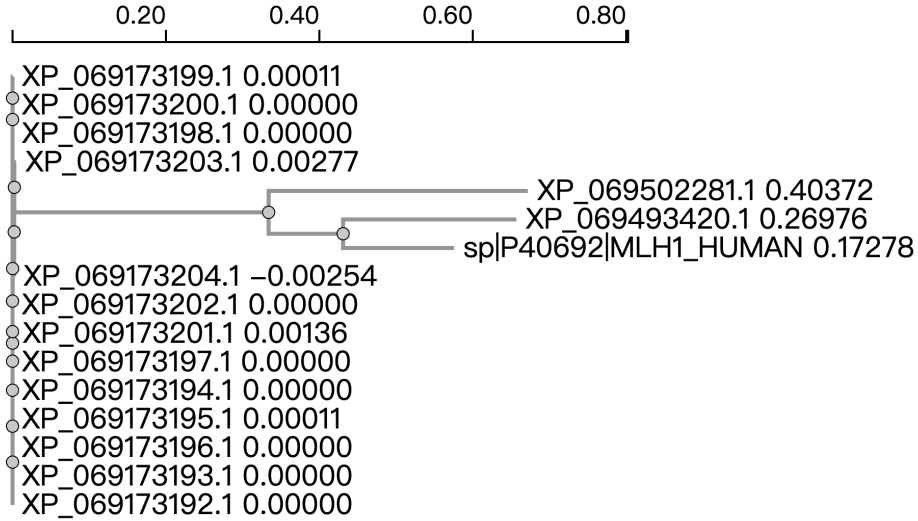


Figure 1: Phylogenetic Tree